Probability and Statistics 1

Course introduction: what this semester is about

OUCHENE Abdelhak – 2026–2027

1 Why statistics and probability?

Modern computer science is built on data that nobody has read in full. A model is trained on a sample and expected to behave on inputs it has never seen; a service receives requests whose arrival cannot be predicted; a security team must decide, from a finite set of logs, whether an anomaly is an attack or a badly configured client; a recommender system compares two variants of its ranking on a fraction of its users before deploying one of them. In each case we observe *a part* — the sample, the logs, the measurements, the A/B test — in order to say something about *a whole* — the population, the future traffic, the threat, the production system — and the observations themselves are affected by variability that no amount of engineering can remove.

Two disciplines answer this need, and they answer it jointly. *Descriptive statistics* condenses a set of observations into a table, a graph and a few numbers, without which the raw data cannot even be read. *Probability* models the variability itself: it describes the set of possible outcomes of an experiment, attaches a degree of belief to each event, and makes it possible to reason about what has not been observed. Neither is optional for an engineer in artificial intelligence. Descriptive statistics without probability produces descriptions that cannot be generalised beyond the sample that produced them; probability without descriptive statistics produces models that are never confronted with data. The two parts of this course are therefore not two subjects but two halves of the same method.

Example.

1. *Model evaluation.* A model obtains an accuracy of 0.87 on a test set of 500 examples. Is 0.87 the accuracy of the model, or the accuracy of the model *on this sample*? Descriptive statistics tells us how to summarise the 500 results; probability tells us how much the number may move on another sample of the same size.

2. *Latency.* The inference latency of a service, measured on 10^4 requests, is reported as a mean of 42 ms. The mean alone is useless to a user who experiences the slowest 1 % of requests; the dispersion parameters and the boxplot of Chapter 2 are what make the report usable.

3. *Security.* An intrusion-detection system flags 200 incidents in a year. Classifying them by attack type, computing the percentages, and identifying the dominant vector is a purely descriptive task; deciding whether the observed increase with respect to last year is significant is a probabilistic one.

4. *A/B testing.* Two ranking algorithms are compared on 2 % of the users. Describing the observed click-through rates is descriptive; concluding that one algorithm is better for the whole user base requires the conditional probabilities and independence of Chapter 3.

2 What statistics is

Definition.

Statistics is the set of methods and techniques used to analyse and process sets of observations, called *data*, in order to facilitate decision-making under conditions of uncertainty. The methods rest on mathematics and make extensive use of computer tools for their implementation.

Example.

Computing the average latency of a service, drawing the histogram of dataset sizes across an organisation, or testing whether two versions of an interface differ are all applications of statistics, in the sense above.

Remark. *Statistics and a statistic*

The word *statistics* denotes the science just defined. A *statistic* is a number computed from a sample: a mean, a percentage, a correlation coefficient. "Statistics is difficult" and "this statistic equals 0.87" use two different words that French, for instance, distinguishes as *la statistique* and *une statistique*.

The discipline has known three broad phases, which explain the tools it uses today. From antiquity to the end of the nineteenth century it remained essentially a set of enumerations, motivated by taxation, demography and insurance. From the nineteenth century to the 1960s, mathematical statistics was constructed as a science by Karl Pearson, William Gosset (Student), Ronald Fisher and Jerzy Neyman, who introduced estimation, hypothesis testing and experimental design. Since the 1960s, and with the development of computing and graphical tools, statistics has undergone a considerable development: the computations that were once reserved for small samples are now performed on millions of observations, and visualisation has become a method of analysis in its own right, not merely a way of presenting results.

3 From data to decision: the statistical approach

Whatever the size of the data and the sophistication of the tools, a statistical study follows the same chain, which is also the plan of the first two chapters.

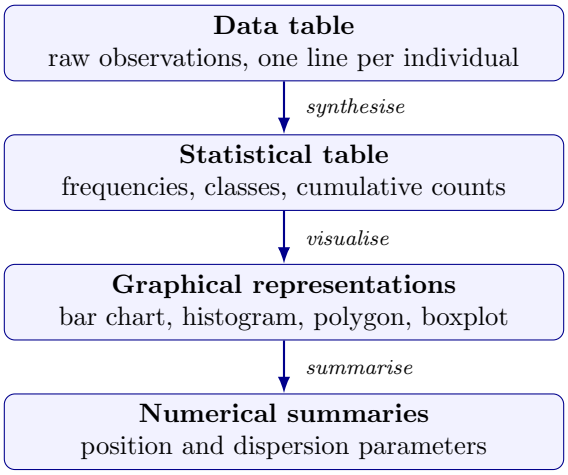


Figure 1: The statistical approach, from raw observations to numerical summaries.

Each stage trades information for legibility. The data table is exact but unreadable; the statistical table is readable but has already lost the identity of the individuals; the graph is immediate but approximate; the numerical summary is a handful of numbers that can be compared, transmitted and tested, but that describe the sample only through the aspects they were designed to capture. The consequence is a rule that governs the whole course: *a summary is always relative to a question.* The mean latency of a service and the latency of its 99th percentile are both legitimate summaries, and they answer different questions; a table that is adequate for one is misleading for the other.

4 The two parts of the course

Definition.

Descriptive statistics is the part of statistics that organises, summarises and describes a set of observations actually collected, without extrapolating beyond them: counting, tabulating, drawing, computing position and dispersion parameters, measuring the association between two characters.

Example.

The frequency table of the code editors used by a cohort of students, the histogram of the sizes of the datasets stored by an organisation, the mean and standard deviation of the inference latency of a model, the correlation between the number of GPUs of a node and its energy consumption are descriptive results: they describe the data at hand, and nothing more.

Definition.

Probability is the part of mathematics that models a random experiment: it defines the set of all possible outcomes, the events that may occur, and a measure of the likelihood of each event. It is the language in which uncertainty is quantified, and the foundation of every inference from a sample to a population.

Example.

1. Throwing a coin, drawing a packet from a network flow, or drawing a user at random from the users of a platform are random experiments; the set of their possible outcomes and the probability of each event are the objects of the probability chapter.

2. The probability that a randomly chosen request is served in less than 50 ms, the probability that a model errs on an input drawn from a given distribution, and the probability that an alert is a false positive given that it was raised by a given rule are all probability computations, and the last one is precisely the kind of question answered by Bayes' formula in Chapter 3.

The course is therefore organised in two parts, each with its own chapter numbering.

Part	Content	Chapters
I. Descriptive statistics	Vocabulary and generalities; one-dimensional series: tables, graphs, position and dispersion parameters	Ch. 1, Ch. 2
II. Probability calculus	Random experiments, events and set algebra, combinatorics; probability spaces and their properties; conditional probability, total probability, Bayes' formula, independence	Ch. 3

Table 1: The two parts of the course.

5 Organisation of the course

The subject is divided into three chapters. The first two form the descriptive part; the third forms the probability part. Each chapter is developed over several sessions, and the exercises of the tutorial sessions follow the progression of the chapters.

Ch.	Title	Content
1	Basic statistical notions and vocabulary	Population, individual, sample; survey, census, poll; character and modalities; qualitative characters (nominal, ordinal) and quantitative characters (discrete, continuous); frequency tables; bar chart, band diagram, pie chart; classes, class centres, amplitude; bar chart for discrete variables, histogram, polygon of frequencies, cumulative frequency curves.
2	Numerical representation of data	Position parameters: mode, arithmetic mean, weighted mean, geometric mean, harmonic mean, quadratic mean, median, quartiles, interquartile range. Dispersion parameters: range, mean absolute deviation, variance, standard deviation, coefficient of variation, boxplot and outliers.
3	Probability calculus	Combinatorics: fundamental principle, arrangements, permutations, combinations. Random experiments, elementary and compound events, realisation of an event, incompatible events, complete system of events, algebra and σ-algebra of events, probabilistic space. Probability spaces: axioms and consequences, conditional probability, independent events, independent experiments. Construction of a probability. Total probability formula, Bayes' formula, composed probabilities.

Table 2: The three chapters of the course and their content.

6 Prerequisites and conventions

The course assumes only elementary mathematics: sets and their operations, algebraic manipulation, summation notation $\sum$, factorials and the first rules of counting. The combinatorics needed in the

probability chapter is recalled in Chapter 3, and no prior knowledge of measure theory is required.

7 Assessment

The final grade is composed of two parts: continuous assessment, which weighs one third of the final grade, and the final examination, which weighs the remaining two thirds.

Component	Detail	Weight
Continuous assessment	Presence: 5 points. Participation in the tutorial sessions: 5 points. Continuous assessment in the tutorial sessions, without prior notification: 10 points. Total of the continuous component: 20 points.	1/3 of the final grade
Final examination	Written examination covering the three chapters.	2/3 of the final grade

Table 3: Assessment scheme.

Remark. *On the continuous assessment in the tutorial sessions*

The 10 points of continuous assessment are obtained through short unannounced evaluations during the tutorial sessions. Their purpose is to verify, session after session, that the definitions and the computational procedures have been acquired, since the material of Chapter 2 and Chapter 3 rests entirely on the vocabulary and the tables of Chapter 1. No notification is given: attendance at the tutorial sessions, and preparation of the exercises before each session, are therefore the only preparation.

8 Further reading

The following references cover the whole course at different levels. They are not required for the examinations, but they are the standard sources on which the notes are based, and they are useful for a student who wishes to go further.

[1] D. Freedman, R. Pisani, R. Purves, *Statistics*, 4th edition, W. W. Norton, 2007. A conceptual introduction to descriptive statistics, with very few prerequisites.

[2] D. C. Montgomery, G. C. Runger, *Applied Statistics and Probability for Engineers*, 7th edition, Wiley, 2018. The reference for engineering curricula; covers Chapters 1 to 3 of this course.

[3] J. L. Devore, *Probability and Statistics for Engineering and the Sciences*, 9th edition, Cengage, 2016. A complete treatment of descriptive statistics and probability, with many exercises.

[4] G. Saporta, *Probabilités, analyse des données et statistique*, 3rd edition, Technip, 2011. The standard French reference, close to the terminology used in these notes.

[5] M. P. Deisenroth, A. A. Faisal, C. S. Ong, *Mathematics for Machine Learning*, Cambridge University Press, 2020. Chapters 6 and 8 give the probabilistic background used in machine learning.

[6] W. Feller, *An Introduction to Probability Theory and Its Applications*, Vol. 1, 3rd edition, Wiley, 1968. The classical reference for the probability part.